

JEE MAIN REVISION MASTER: COMPOUNDS CONTAINING NITROGEN 2026 3 QUESTIONS 5% WEIGHTAGE

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 27 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Kcet Compounds Containing Nitrogen - Core Application	27 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Kcet Compounds Containing Nitrogen - Core Application (Very High Yield — 27 PYQs)

Core Principle: Key problem pattern in Kcet Compounds Containing Nitrogen.

■ **Caution / JEE Pitfall:** Watch for algebraic and sign errors.

■ **Speed Trick:** Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "Organic compound 'D' is..." [Ans: N/A]
- [2026 Shift] Q#2 (MCQ): "Basic strength of alkylamines in aqueous phase is not decided by _____." [Ans: N/A]
- [2026 Shift] Q#3 (MCQ): "Nitration of aniline in strong acidic medium gives significant amount of m-nitroaniline because..." [Ans: B]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

Organic compound 'D' is

- (A)
- (B) CH_3COOH
- (C) CH_3CHO
- (D)

Correct Answer: None

Solution Analysis: Benzene first reacts with concentrated HNO_3 and concentrated H_2SO_4 at 323-333K . This is nitration of benzene, and the product formed is nitrobenzene, marked as (A). $\text{C}_6\text{H}_6 \xrightarrow{\text{Conc. HNO}_3/\text{Conc. H}_2\text{SO}_4} \text{C}_6\text{H}_5\text{NO}_2$ (A) Nitrobenzene (A) is then reduced using Fe/HCl . The nitro group, $-\text{NO}_2$, changes into an amino group, $-\text{NH}_2$. So the product is aniline, marked as (B).

$\text{C}_6\text{H}_5\text{NO}_2 \xrightarrow{\text{Fe}/\text{HCl}} \text{C}_6\text{H}_5\text{NH}_2$ (B)